

FLORES AP, Portugal

WMO: 085010

Lat: 39.4572N

Lon: 31.1308W

Elev: 29

StdP: 100.98

Time Zone: -1.00 (AZR)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	10.2	3.3	4.8	11.0	4.2	5.1	11.6	14.2	14.7	13.1	15.7	4.6	240	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	26.9	22.4	26.1	22.1	25.2	21.6	23.2	25.5	22.7	25.1	22.2	24.6	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.0	24.4	21.9	16.6	24.3	21.2	16.0	23.9	68.4	25.6	66.8	25.3	65.1	24.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.8	9.5	DB	7.5	28.9	0.7	0.8	7.0	29.5	6.6	29.9	6.2	30.4	5.6	30.9	(4)
(5)			WB	5.2	24.4	0.7	0.6	4.7	24.8	4.3	25.1	4.0	25.5	3.5	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	15.0	14.7	14.7	15.5	17.1	19.3	21.9	23.0	21.8	19.4	17.4	15.5	(6)
(7)		DBStd	3.44	1.92	1.84	1.76	1.64	1.77	1.79	1.70	1.52	1.62	1.99	2.03	2.08	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	604	104	103	111	86	48	11	1	0	1	12	42	88	(9)
(10)		CDD10.0	2906	155	131	147	166	219	278	368	404	354	292	221	172	(10)
(11)		CDD18.3	468	1	0	0	1	9	39	110	146	104	45	12	2	(11)
(12)		CDH23.3	1007	0	0	0	1	5	18	261	516	187	19	1	0	(12)
(13)		CDH26.7	29	0	0	0	0	1	0	5	21	2	0	0	0	(13)
(14)	Wind	WSAvg	4.4	5.1	5.0	5.2	4.8	4.0	3.8	3.1	3.2	3.9	4.4	4.6	5.2	(14)
(15)	Precipitation	PrecAvg	1564	173	151	152	99	98	93	67	71	119	181	156	205	(15)
(16)		PrecMax	2067	322	356	380	303	266	241	157	159	226	405	259	439	(16)
(17)		PrecMin	972	57	21	48	3	22	11	10	13	43	29	40	46	(17)
(18)		PrecStd	261	76	69	82	62	61	53	37	39	52	92	59	95	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	19.2	19.2	20.9	23.1	25.0	27.2	28.1	26.9	24.9	23.1	20.8	(19)
(20)			MCWB	17.0	16.2	16.4	16.9	19.0	21.0	22.5	22.6	22.4	21.2	19.4	18.1	(20)
(21)		2%	DB	19.1	18.4	18.4	19.6	21.7	23.8	26.2	27.2	25.8	23.9	21.9	19.9	(21)
(22)			MCWB	16.8	16.0	16.0	16.4	18.3	20.3	22.0	22.5	22.0	20.8	19.0	17.5	(22)
(23)		5%	DB	18.5	18.0	17.9	18.9	20.8	22.9	25.6	26.3	25.1	23.1	21.0	19.2	(23)
(24)			MCWB	16.3	15.8	15.7	16.2	17.8	19.8	21.7	22.3	21.7	20.3	18.6	17.0	(24)
(25)		10%	DB	17.9	17.2	17.2	18.1	19.9	22.1	24.9	25.8	24.3	22.4	20.2	18.7	(25)
(26)			MCWB	15.9	15.3	15.2	15.7	17.2	19.3	21.2	22.1	21.3	19.9	18.1	16.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	17.6	17.6	17.9	19.7	21.9	23.5	24.0	23.6	22.2	20.5	18.8	(27)
(28)			MCDWB	19.0	18.3	18.3	19.4	21.5	23.7	25.8	26.2	25.3	23.7	21.8	20.1	(28)
(29)		2%	WB	17.1	16.7	16.7	17.2	19.0	21.0	22.8	23.2	22.9	21.5	19.7	18.0	(29)
(30)			MCDWB	18.5	17.8	17.9	18.7	20.9	22.9	25.2	25.9	24.7	23.1	21.0	19.3	(30)
(31)		5%	WB	16.7	16.2	16.1	16.7	18.3	20.4	22.2	22.9	22.4	20.9	19.0	17.5	(31)
(32)			MCDWB	18.1	17.4	17.4	18.2	20.3	22.3	24.7	25.6	24.3	22.7	20.5	18.8	(32)
(33)		10%	WB	16.2	15.7	15.6	16.1	17.6	19.7	21.7	22.4	21.9	20.4	18.5	16.9	(33)
(34)			MCDWB	17.7	17.1	17.0	17.7	19.6	21.7	24.2	25.1	23.9	22.2	20.0	18.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.7	4.0	3.8	4.1	4.5	4.5	4.9	4.4	4.1	3.5	3.9	(35)	
(36)			MCDBR	4.0	4.2	4.1	4.9	5.4	5.1	5.5	5.5	4.9	4.4	4.1	4.1	(36)
(37)			MCWBR	3.7	3.9	3.7	3.9	3.8	3.6	3.4	3.4	3.4	3.4	3.4	4.0	(37)
(38)		5% WB	MCDWB	3.6	3.8	3.6	4.4	5.0	4.5	5.0	5.1	4.3	4.0	3.5	3.7	(38)
(39)			MCWBR	3.8	4.0	3.6	3.9	3.9	3.6	3.6	3.6	3.4	3.6	3.5	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.358	0.370	0.399	0.392	0.386	0.386	0.380	0.376	0.375	0.368	0.361	0.362	(40)	
(41)		taud	2.476	2.452	2.352	2.411	2.457	2.493	2.532	2.558	2.554	2.552	2.511	2.430	(41)	
(42)		Ebn at Noon	804	846	855	885	893	891	893	890	871	840	796	764	(42)	
(43)		Edh at Noon	81	94	114	114	111	107	102	97	92	84	77	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.90	2.73	3.81	5.08	5.94	6.28	6.58	6.12	4.71	3.27	2.15	1.65	(44)	
(45)		RadStd	0.17	0.16	0.27	0.44	0.42	0.48	0.37	0.28	0.21	0.18	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page